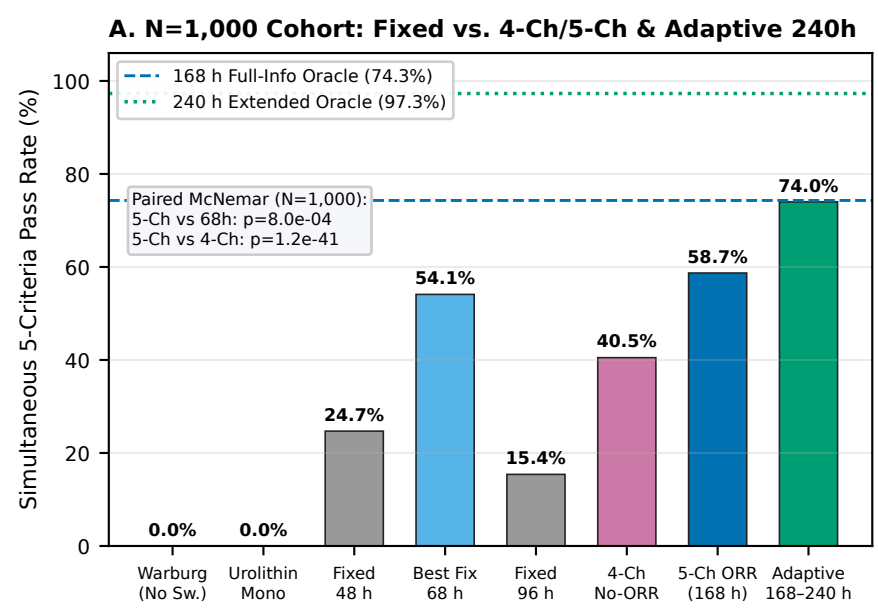
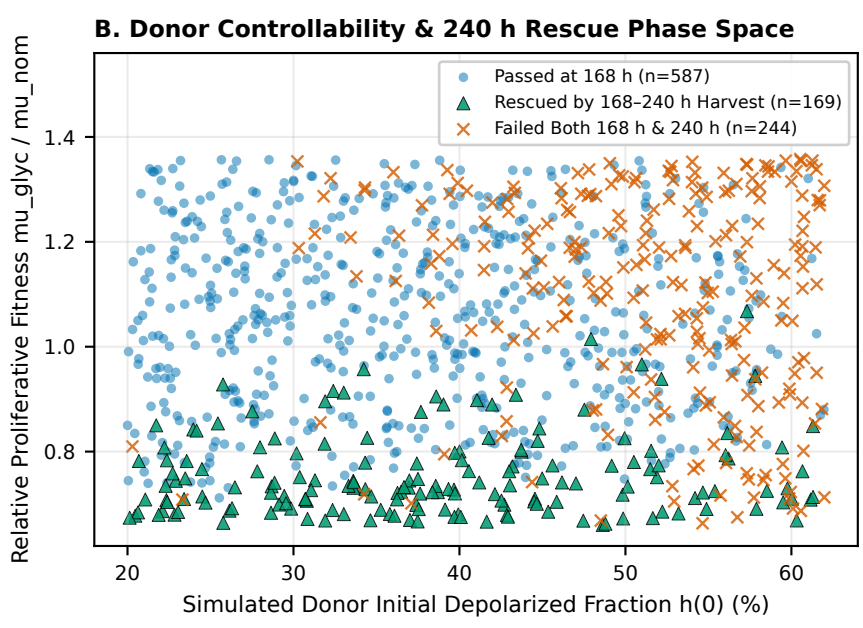
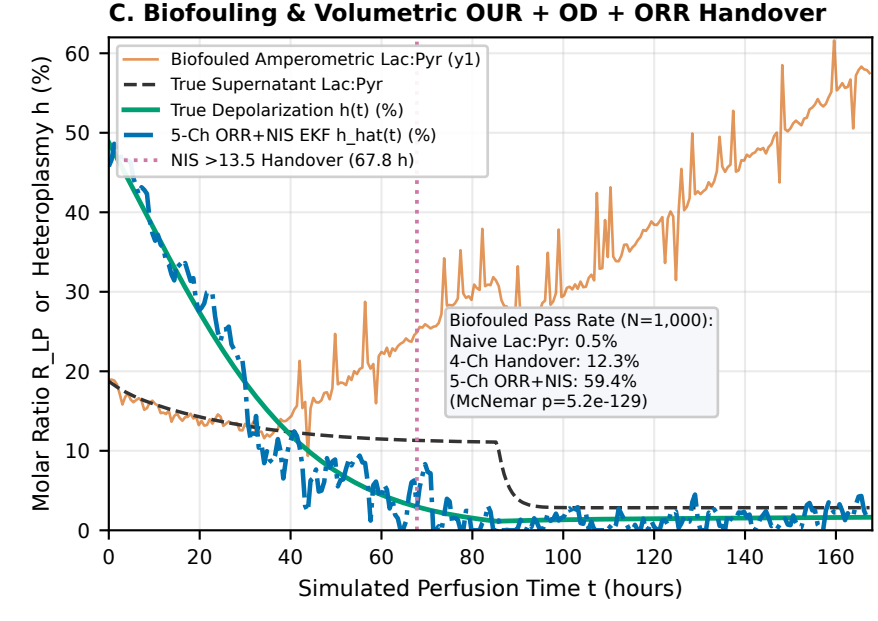
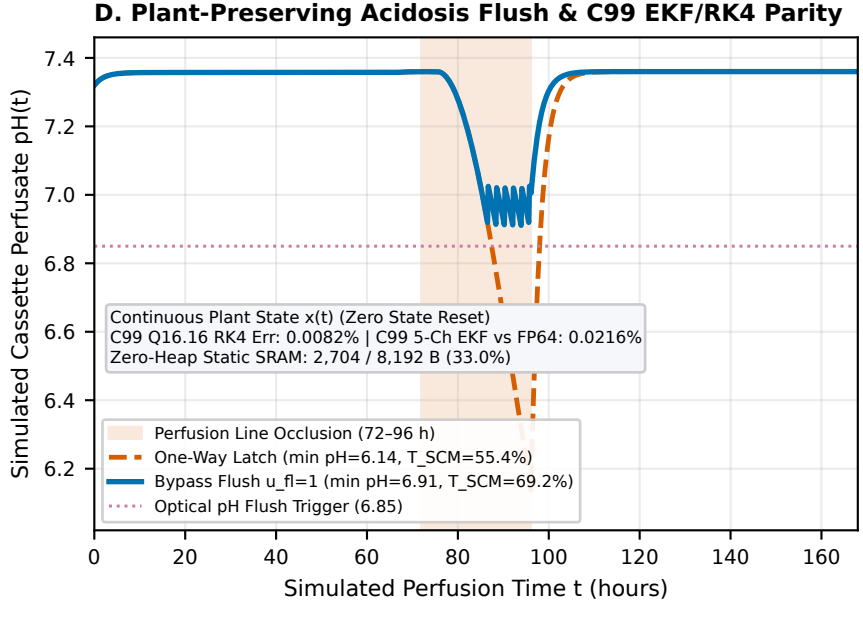

A. N=1,000 Cohort: Fixed vs. 4-Ch/5-Ch & Adaptive 240h
168 h Full-Info Oracle (74.3%)
240 h Extended Oracle (97.3%)
Paired McNemar (N=1,000):
5-Ch vs 68h: p=8.0e-04
5-Ch vs 4-Ch: p=1.2e-41
Simultaneous 5-Criteria Pass Rate (%)
0.0%
0.0%
24.7%
54.1%
15.4%
40.5%
58.7%
74.0%
Warburg (No Sw.)
Urolithin Mono
Fixed 48 h
Best Fix 68 h
Fixed 96 h
4-Ch No-ORR
5-Ch ORR (168 h)
Adaptive 168–240 h


B. Donor Controllability & 240 h Rescue Phase Space
Passed at 168 h (n=587)
Rescued by 168–240 h Harvest (n=169)
Failed Both 168 h & 240 h (n=244)
Relative Proliferative Fitness mu_glyc / mu_nom
Simulated Donor Initial Depolarized Fraction h(0) (%)


C. Biofouling & Volumetric OUR + OD + ORR Handover
Biofouled Amperometric Lac:Pyr (y1)
True Supernatant Lac:Pyr
True Depolarization h(t) (%)
5-Ch ORR+NIS EKF h_hat(t) (%)
NIS >13.5 Handover (67.8 h)
Biofouled Pass Rate (N=1,000):
Naive Lac:Pyr: 0.5%
4-Ch Handover: 12.3%
5-Ch ORR+NIS: 59.4%
(McNemar p=5.2e-129)
Molar Ratio R_LP or Heteroplasmy h (%)
Simulated Perfusion Time t (hours)


D. Plant-Preserving Acidosis Flush & C99 EKF/RK4 Parity
Continuous Plant State x(t) (Zero State Reset)
C99 Q16.16 RK4 Err: 0.0082% | C99 5-Ch EKF vs FP64: 0.0216%
Zero-Heap Static SRAM: 2,704 / 8,192 B (33.0%)
Perfusion Line Occlusion (72–96 h)
One-Way Latch (min pH=6.14, T_SCM=55.4%)
Bypass Flush u_fl=1 (min pH=6.91, T_SCM=69.2%)
Optical pH Flush Trigger (6.85)
Simulated Cassette Perfusate pH(t)
Simulated Perfusion Time t (hours)